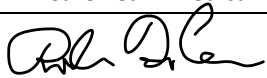


Guideline Title: Emergency Blood Product Administration

Guideline Number: GUID-3203-PR025

Issue date: 7/10/2019	Replaces Dept. Policy:
Revision dates: 07/16/2026	Developed by: Air Care Team
Approved by: Dr. Danielle DiCesare, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** This department guideline is used to provide safe and effective administration of uncrossmatched blood and blood products to Air Care Team patients with hemorrhagic shock from trauma or other mechanisms. This guideline is also used to recognize and effectively treat transfusion reactions.
- II. **DEFINITIONS:**
 - A. Blood products: liquid plasma A and red blood cells, O positive and O negative.
 - B. IV – Intravenous
 - C. SBP – Systolic Blood Pressure
 - D. NS – Normal Saline
- III. **DEPARTMENT GUIDELINE:**
 - A. **EQUIPMENT**
 - i. Personal protective equipment (PPE).
 - ii. Appropriate blood product.
 - iii. Filtered blood tubing or leukocyte reduction filter.
 - iv. A patent IV catheter with Normal Saline infusing.
 - v. Pressure bags may be utilized with caution for transfusing blood as long as the pressure inside the bag does not exceed 300 mmHg.
 - B. **INDICATIONS**
 - i. **Adult** Indications
 1. Blunt or penetrating trauma, hypovolemia with acute blood loss not limited to gastrointestinal hemorrhage, post-partum hemorrhage, or severe coagulopathy, signs and symptoms of hemorrhagic shock with:
 - a. SBP < 90 mmHg
 - b. Pulse rate > 110 bpm
 - c. Evidence of peripheral vasoconstriction including cool, pale skin and delayed capillary refill > 2 seconds.
 2. Hemorrhage with documented HGB < 7 g/dL

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ii. **Pediatric Indications**

1. Blunt or penetrating trauma, hypovolemia with acute blood loss, signs and symptoms of hemorrhagic shock with:
 - a. Evidence of hypoperfusion (pallor, peripheral vasoconstriction including cool, pale skin and delayed capillary refill > 2 seconds)
 - b. SBP < (70 + 2 x age)
 - c. Pulse Rate – clinical context should be taken into account when assessing pulse rate number (e.g. presence of fever, dehydration, anxiety, pain, etc.)
 - i. Neonates (less than 28 days) < 100 bpm or > 180 bpm
 - ii. Infant (1 month – 1 year) < 80 bpm or > 160 bpm
 - iii. Child (1 – 10 years) < 60 bpm or > 130 bpm
 - iv. Older Child (>10 years) < 60 bpm or > 110 bpm
2. Hemorrhage with documented HGB <7 g/dL

C. CONTRAINDICATIONS

1. Patients or patient's legal representatives' refusal of blood or blood products due to religious or personal reasons.
2. Previous history of significant blood transfusion reaction.

D. DOSAGE

i. **ADULT**

1. **Bolus entire unit of liquid plasma A and PRBC IV/IO**

ii. **PEDIATRIC**

1. **PRBC's: 10 ml/kg IV/IO.** Consider a fluid bolus of 20 ml/kg of 0.9% NS IV/IO per *GUID3203-PED009 - Pediatric Hypotension* before transfusion.
2. **Liquid Plasma A: Liquid plasma infusion at 10 ml/kg** and infused at a maximum rate of 0.5 ml/kg/min.

E. PROCEDURE:

- i. Consider the administration of Tranexamic Acid (TXA), per *GUID3203-T016 Tranexamic Acid for Severe Hemorrhage in Trauma Patients*.
- ii. If patient has a central venous line or intraosseous line, consider **Calcium Chloride 1 G slow IV/IO push** after every 2 units of blood have been administered, up to 3 G total. This medication should not be given through peripheral IV access and can be delayed until patient has been transferred to receiving center.
- iii. Utilize appropriate PPE.
- iv. Initially assess the following:
 1. Vital signs.
 2. Skin condition; i.e. rashes, redness, etc.
 3. Size and integrity of IV site.
 - a. Adults: Prefer 18 g for PRBCs and 20 g for other products.
 - b. Pediatrics: 22-24 g.

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4. Never transfuse blood in piggyback fashion into any IV with medications infusing. Only piggyback into NSS Line.
5. Breath sounds.
6. Fluid volume status.
7. History of previous transfusion reactions.
- v. Blood products must be maintained at a controlled temperature of 1-6 degrees Celsius during transport and must be infused within 4 hours of removal from the thermal control environment.
- vi. Select the appropriate blood product
 1. The preferred order of blood products administration:
 - a. One (1) IV available:
 - i. Adult: Liquid Plasma, PRBC, Liquid Plasma, and PRBC.
 - ii. Pediatric: PRBC, then Liquid plasma
 - b. Two (2) IV's available:
 - i. IV #1 – 1 unit of Plasma set up on blood tubing with NS
 - ii. IV #2 – 1 unit of PRBC's set up on blood tubing with NS
 2. Type A liquid Plasma for both males and females of all ages.
 3. Males: O positive (O+) Packed red blood cells (PRBC's)
 4. Females:
 - a. Childbearing age or younger
 - i. O negative (O-) PRBC's if available.
 - ii. O+ can be given to women of childbearing years in life-threatening situations if O- is not available: notify the receiving facility that patient received O+ and will require rho(d) immune globulin (RhoGAM).
 - b. Beyond childbearing age (>50 years old): O+ PRBC's
- vii. Blood should be warmed during administration, if possible. See *GUID3203-PR026 – Warming of Blood Products and Intravenous Fluids*.
- viii. Once the patient has been identified as meeting the criteria for emergent transfusion of unmatched blood, complete the double verification *Bedside Verification Process for Emergency Release Blood* based on Orlando Health Policy #2075 *Blood/Blood Product Administration: Routine and Emergent*.
 1. At the bedside the Transfusionist (ACT RN) or Independent Verifier (Credentialed ACT Paramedic) verifies the following information:
 - a. patient name/alias if available
 - b. medical record (MR)# if available
 - c. blood product label type (O Negative or O Positive),
 - d. the blood tag unit number and expiration date.
 - e. Inspect blood and color tab for any changes.

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2. Enter date and time on the designated line on the blood tag if all information matches
3. After verification by both the Transfusionist (ACT RN) and the Independent Verifier (Credentialed ACT Paramedic); the blood is transfused.
- ix. May place blood product in pressure bag and inflate the bag not to exceed 300 mmHg.
- x. Establish first unit of liquid plasma set up on blood tubing with NS, once completed, flush line with NS and hang 1 unit of PRBC's, transfuse in similar manner. If after reassessment, patient still meets criteria and time allows, team may transfuse another unit of Plasma followed by PRBC for total of two units PRBC and two units of plasma.
- xi. Between each transfusion:
 1. Flush IV with NS flush.
 2. Record time of completion on tag
- xii. Reassess airway patency, hemodynamic stability and treat any transfusion reaction or complications per *GUID3203-M004 – Anaphylaxis* or *GUID3203-PED004 – Pediatric Allergic Reaction/Anaphylaxis*.
 1. Assess and record the patient's vital signs, IV site condition, and any signs and symptoms of transfusion reaction (e.g. fever, chills, nausea, pain, and change in vital signs or respiratory distress).
 2. If a transfusion reaction occurs:
 - a. Stop transfusion immediately.
 - b. Assess vital signs.
 - c. Remove blood bag and tubing and set aside.
 - d. Assure bag and tubing are given to receiving RN/appropriate medical staff/blood bank.
 - e. Provide systemic support as required.
- xiii. Once transfusion of any products is complete:
 1. Complete the Air Care Team *Uncrossmatched Emergency Blood Release Form* in its entirety.
 2. Notify receiving physician of any transfusion reactions and treatment provided.
 3. Notify receiving physician if O+ blood was given to a female of childbearing age to ensure patient receives adequate rho (D) immune globulin.
 4. Obtain all necessary signatures.
 5. Complete tags on blood/plasma in their entirety.
 6. Leave "segments" attached to each unit of blood with receiving facility nurse to be given to blood bank for crossmatch.
 7. Leave a copy of the *Uncrossmatched Emergency Blood Release Form* with receiving facility, original form gets scanned in and attached to Air Care Team chart.
 8. If patient brought to Orlando Regional Medical Center, save blood bags and bring to blood bank with the *Uncrossmatched Emergency Blood Release Form*.

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xiv. Restocking Blood Products

1. Resupply at Orlando Regional Medical Center (ORMC) if able.
2. If unable to restock blood products at ORMC, contact OneBlood for resupply. Just FYI, OneBlood will resupply but it could be multiple hours and they do need someone (medical crew member, pilot, or mechanic) available to sign for the product.

IV. **DOCUMENTATION:**

A. EMS Charts

- i. Blood Product Administration on page 7 to include
 1. Presence or absence of transfusion reaction
- ii. Patient Summary – Page 8
 1. Vital signs before transfusion and when completed

B. Air Care Team Uncrossmatched Emergency Blood Release Form

C. OneBlood tags on individual units

- D. Complete the *Blood Component Disposition Log* immediately upon returning to base and email to OneBloodAirMedicalTeam@oneblood.org and retain a copy at the base in the Blood Product Tracking binder.

V. **REFERENCES:**